

MAN00035C

Quantitative Analysis in Business Studies

2026 – Week 2



UNIVERSITY
of York

School for Business
and Society



Week 2: Probability

Reminder

- Revise the lecture material.
- Try as many times as you would like the formative tests - stop only when you feel confident.
- Formative tests do not feed into your marks and your results are not monitored by the teaching team for the purpose of marking.
- Post your questions on the weekly VLE 'Discussion Board'

Reminder

- All the information is available on VLE in the module handbook
 - **1200 word assignment (report/course work)**, due in **Week 11** (date to be announced soon on the VLE site) counting for **50%** of the module mark. The assignment will be released to you at **the end of Week 4**. The assignment is to be your own work (do not approach the teaching staff for answers).
 - **A closed exam** due in the Semester 2 **Common Assessment Period** (date to be announced later on the VLE site) counting for **50%** of the module mark.
- There are 3 computer practicals in Weeks 9, 10 and 11. You can find the details in your timetable.

Structure of today's session

We're going to look at

- what are probabilities?
- how to calculate probabilities?
- how to combine probabilities?

Reading: Chapter 6 in Oakshott

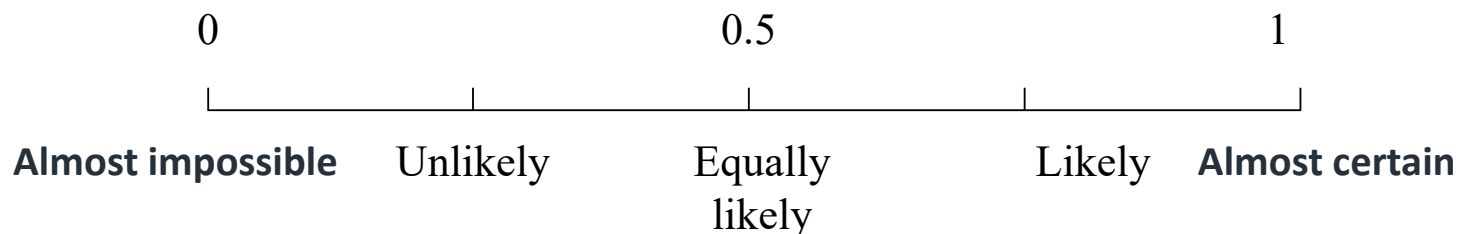
Car insurance

- Imagine you own a £40K car
- Assuming that the probability of it being stolen is 1/1000
- Then the expected loss for the insurance company is $£40,000 \times 1/1000 = £40$
- Annual premium for theft on your insurance must be more than £40

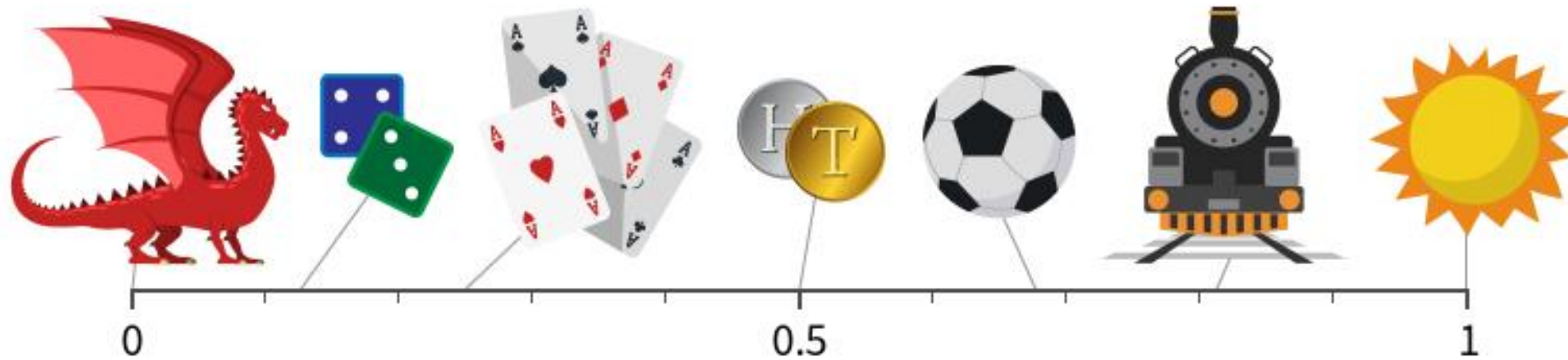


Probability

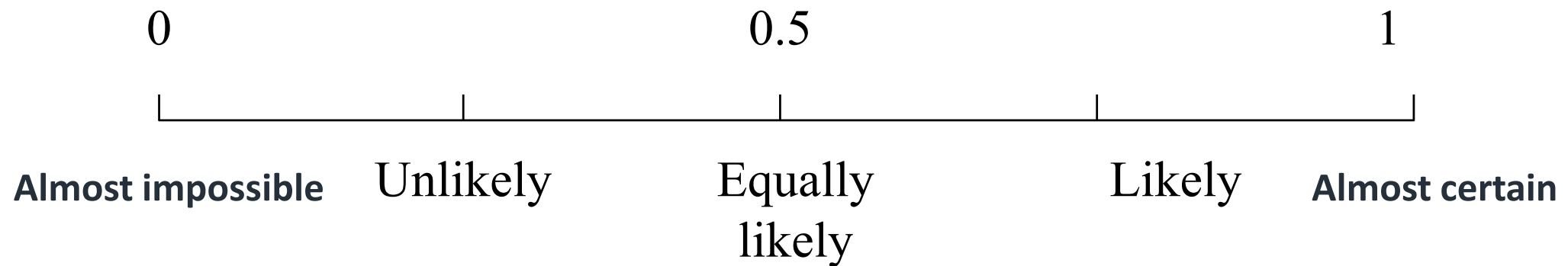
- The probability of an event is how likely it is to happen
- Some things are **almost certain**. They have a probability of **1**.
- Some things are **almost impossible**. They have a probability of **0**.
- Some things are somewhere in the middle. They have **probability between 0 and 1**.
- Probability is measured on a sliding scale from **0 (almost impossible) to 1 (almost certain)**



Example



Source: <https://mathigon.org/course/probability/introduction>



Probability

How to allocate probabilities to events?

1. Classical ('a priori' or objective) approach

Events assigned (sometimes equal) probabilities based on theory

2. Empirical or relative frequency approach

Probabilities retroactively assigned through experiments

3. Subjective approach

Quantification of personal belief.

E.g. performance of a new technology startup firm

The classical or 'a priori' approach

$$P(A) = \frac{\text{number of outcomes representing event A}}{\text{total number of possible outcomes}}$$

- where all of the outcomes are assumed to be equally likely.

Example

What is the probability of obtaining an odd number score on one throw of a die?

$$P(\text{odd number score}) = \frac{3}{6} = 0.5$$

The empirical or relative frequency approach

Requires **past**, or **experimental data**:

$$P(A) = \frac{\text{number of times event A occurred}}{\text{total number of observations}}$$

Example

Out of 200 light bulbs manufactured on a particular day 15 were faulty,

$$P(\text{light bulb manufactured on given day is faulty}) = \frac{15}{200} = 0.075$$

The subjective approach

A person uses judgement to assess the probability.

Example

Consider a new product that does not exist on the market.

"I estimate that there is a 0.9 probability that this product will make a profit next year."

Probability and Percentages

Probabilities and percentages are **similar concepts** in some ways.

You can also measure probability in percentages, but then it's a percentage chance, not a probability.

(Probability of 0.5, the percentage chance of 50%)

- ❖ If a question asks for a probability, it's looking for a value between 0 and 1.
 - ❖ E.g. 0.5 probability of getting Head on a coin flip
- ❖ If it asks for a percentage chance, it wants a value between 0% and 100%.
 - ❖ E.g. 50% discount on a product for sale

Independent events

- Two events A and B are **independent** if the probability of A occurring is not affected by the occurrence of B and vice versa.

Example

- Consider an observation (rained in Bristol, it is a weekday)
- $P(\text{it will rain in Bristol on a certain day})$
- is independent of** whether that day is a weekday or not,
 - but is **not independent of** what month that day occurs in, as some months are rainier than others.

Conditional Probability

This is denoted by $P(A \mid B)$ which means "the probability of A occurring *given that* B has occurred".

Example

A supermarket manager might judge:

$P(\text{delivered goods are in good condition}) = 0.9$

but

$P(\text{delivered goods are in good condition} \mid \text{packaging is damaged}) = 0.3$

The multiplication rule

This gives the probability of one event *and* another event occurring.

If the **events are independent** then:

$$P(A \text{ and } B) = P(A) \times P(B)$$

The multiplication rule cont.

Example

If you toss a coin twice then the probability of getting two heads is:

$$= 0.5 \times 0.5 = 0.25$$

Since the two events are **independent**

The multiplication rule cont.

Quality of your password:

- If you have a 6-digit password using only numerical digits you have $10 \times 10 \times 10 \times 10 \times 10 \times 10$ combinations which equals $10^6 = 1,000,000$, which sounds like a lot of possibilities but not for a computer.
- If you include letters then the 6 digits have now 36 combinations (26 letters and 10 digits), then the number of combinations is 36^6 which is over 2 billion.



The multiplication rule cont.

If two events are **dependent** then:

$$P(A \text{ and } B) = P(A) \times P(B \mid A)$$

Example

A box contains 100 spark plugs of which 20 are defective.

If two plugs are to be sampled from the box *without replacement* what is the probability that they will both be defective?

$$\begin{aligned} &P(\text{first and second plugs defective}) \\ &= P(\text{first plug defective}) \\ &\quad \times P(\text{second plug defective} \mid \text{first defective}) \\ &= 20/100 \times 19/99 \\ &= \mathbf{0.0384} \end{aligned}$$

The addition rule

This gives the probability of either one event or another occurring.

$$P(\text{A or B}) = P(\text{A}) + P(\text{B}) - P(\text{A and B})$$

If the events are **mutually exclusive** (cannot both simultaneously happen), then:

$$P(\text{A and B})=0 \quad \Rightarrow \quad P(\text{A or B}) = P(\text{A}) + P(\text{B})$$

The addition rule cont.

Example

One card is drawn from a well shuffled pack. Find the probability that it is **(i) either an ace or a jack; (ii) either a king or a diamond.**

(i) The events "ace" and "jack" are mutually exclusive so:

$$P(\text{ace or jack}) = 4/52 + 4/52 = 8/52$$

(ii) The events "king" and "diamond" are not mutually exclusive so:

$$P(\text{king or diamond}) = 4/52 + 13/52 - 1/52 = 16/52$$

Collectively exhaustive events

- Two or more events are said to be **collectively exhaustive** if they cover all possible outcomes
- If a set of events is both **mutually exclusive** and **collectively exhaustive**, it means that exactly one (and only one) of them must happen
- Hence: If events **A** and **B** are **mutually exclusive** and **collectively exhaustive** then

$$P(A) = 1 - P(B)$$

Conclusions

- what are probabilities?
- how to calculate probabilities?
- how to combine probabilities?

Next we will consider:

- **Bayes' rule** and applications
 - **Expectations**
 - Counting Arguments
- Reading: Chapter 6 Probabilities

Bayes' Rule



Probability: Marbles in a bag

- We are all familiar with probability to some extent.
- Imagine **there's a bag of 6 green marbles**. We take a marble out of the bag (blind sampling).
- We can agree that the probability of that marble being green (written **$P(\text{Green})$**) is **1**.
- And that **$P(\text{Blue})$** is **0**?
- (**$P(\text{Green})$** refers to the Probability of the event "Green". **P** is not a number.)

Probability: Marbles in a bag (2)

- Now consider a bag with **3 Green** marbles and **3 Blue** ones (overall 6 marbles).
- Can we agree $P(\text{"Green"}) = P(\text{"Blue"}) = 3/6 = 0.5$?
- We often like to show probabilities in natural frequencies as it is easier to see where they are coming from.

Probability: Marbles in a bag (3)

- What about a second draw from our bag of 3 Green and 3 Blue marbles?

(Without replacement)

- What's the probability of drawing a Green marble second?

We don't know!

- It matters what we got *first*, and is either:

$\frac{2}{5}$ (if we got Green first) or $\frac{3}{5}$ (if we got Blue first).

- In **neither** case is it the $\frac{3}{6}$ we had!

Conditional Probability

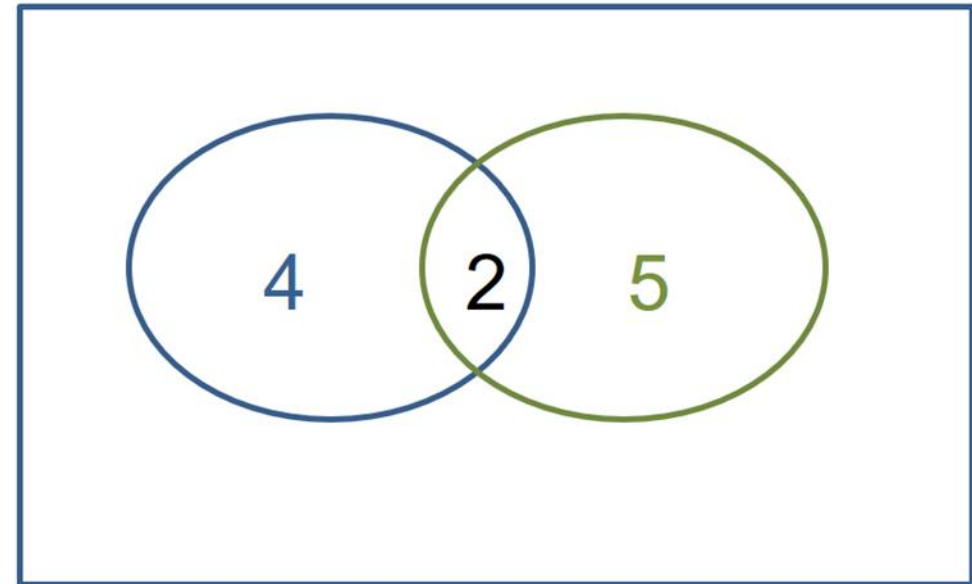
- This idea that probabilities depend on previous events is that of **conditional** probability.
- It works just like “normal” probability, but you restrict the “possible results” in some way.
- For our 3 and 3 bag, we could write:

$$P(G \text{ 2nd} \mid B \text{ 1st}) = \frac{3}{5}, \text{ and } P(G \text{ 2nd} \mid G \text{ 1st}) = \frac{2}{5}.$$

(To be read “the probability of Green second given (or if) we got Blue first is 3/5” and “the probability of Green second given (or if) we got Green first is 2/5”.)

Diagrams can help

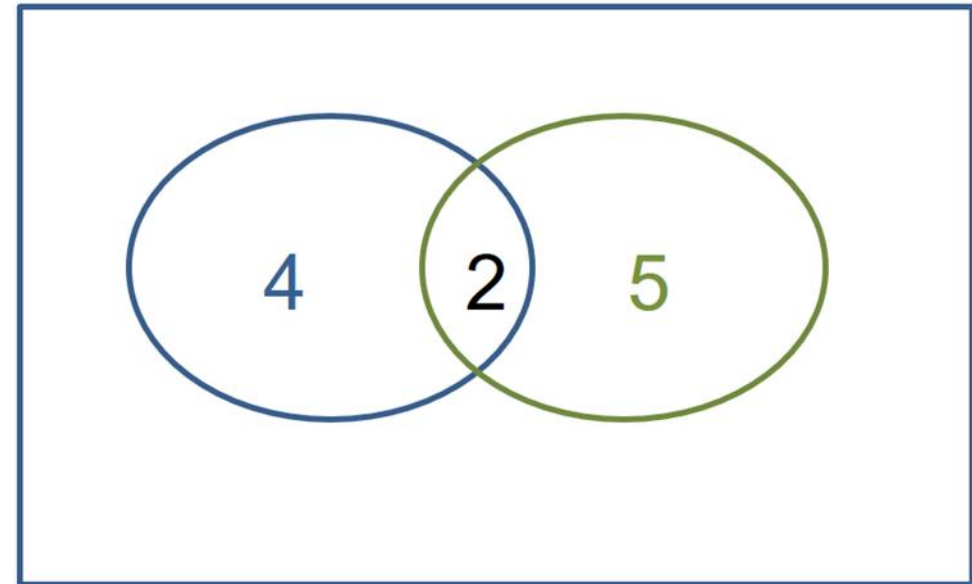
- What if we had multicoloured marbles?
 - 4 that are just Blue,
 - 5 that are just Green, and
 - 2 that are both!



Diagrams can help (2)

- What if we had multicoloured marbles?

$$\begin{aligned}P(B) &= \\P(G) &= \\P(B \text{ and } G) &= \end{aligned}$$



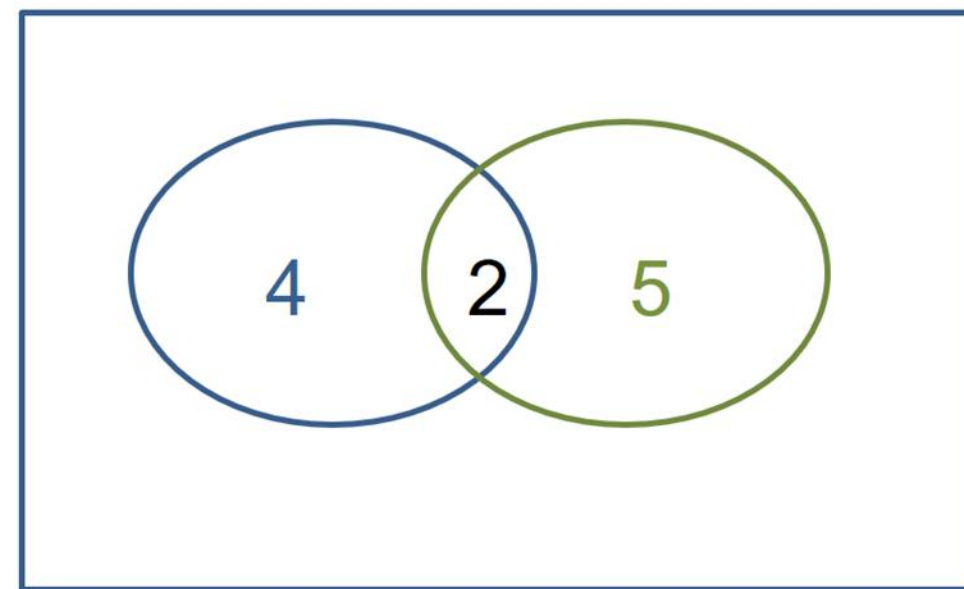
Diagrams can help (2)

- What if we had multicoloured marbles?

$$P(B) = \frac{6}{11}$$

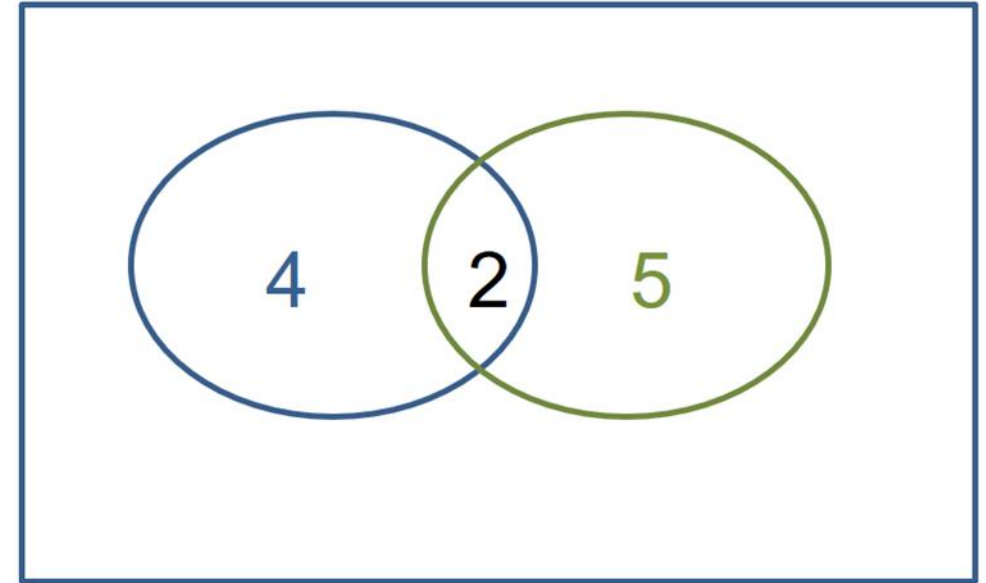
$$P(G) = \frac{7}{11}$$

$$P(B \text{ and } G) = \frac{2}{11}$$



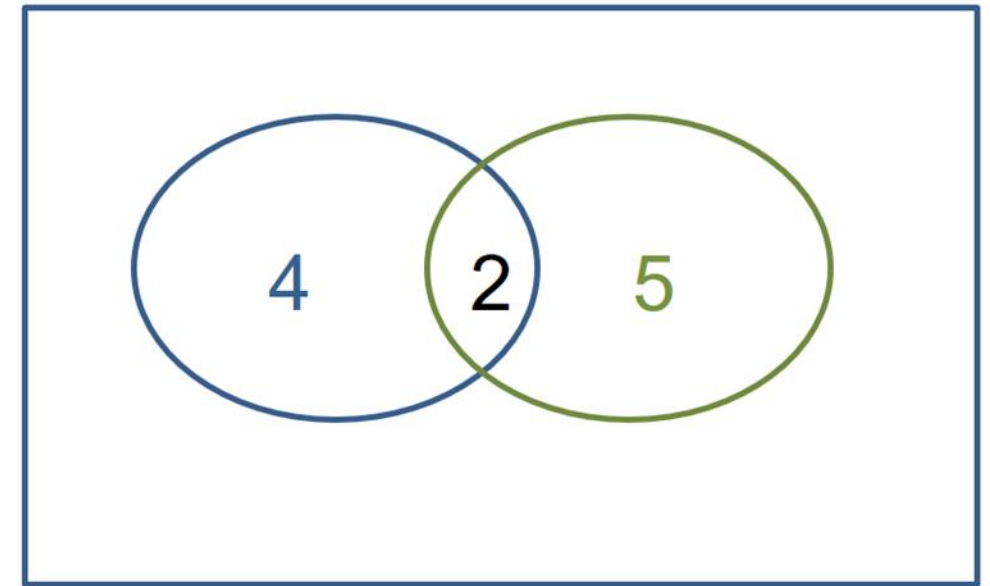
Diagrams can help (3)

- What if, when you were pulling a marble out of the bag, the sun got in your eyes?
- And your friend says “you got a **Blue one!**”.
- You reply “but I wanted at least some **Green** on it!”
- *What’s the probability of getting a Green one, given that we know it’s blue too?*



Diagrams can help (4)

- We know it's **Blue**.
- We want $P(\text{"Green"} \mid \text{"Blue"})$.
- We know we're in the 4+2 set of marbles (those are the ones that are Blue),
 - and want one of the 2 (those are the ones that are also Green).



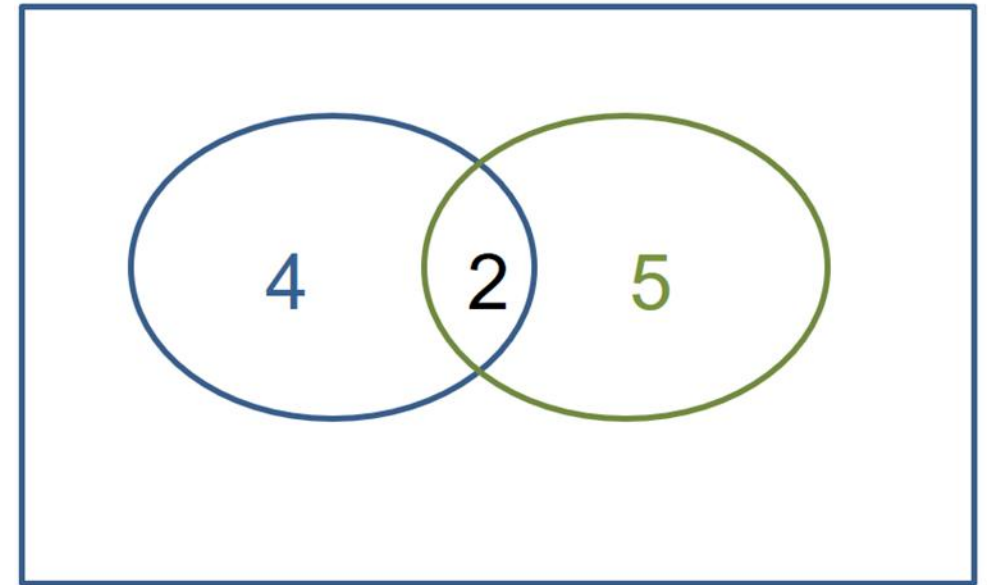
Diagrams can help (5)

Quite reasonably, therefore:

$$P(G|B) = \frac{2}{6}.$$

And if we think about where these numbers are coming from, we have:

$$P(G|B) = \frac{P(G \text{ and } B)}{P(B)}$$



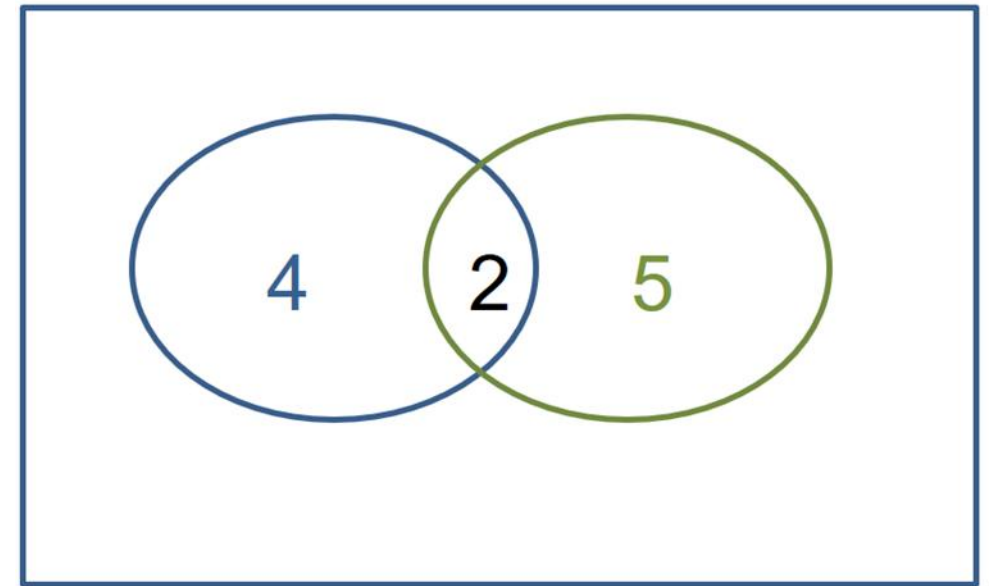
Diagrams can help (6)

Note that $P(G|B)$ and $P(B|G)$ might look very similar.

That doesn't mean they will have similar values though!

$$P(B|G) = \frac{2}{7} \text{ while } P(G|B) = \frac{2}{6}$$

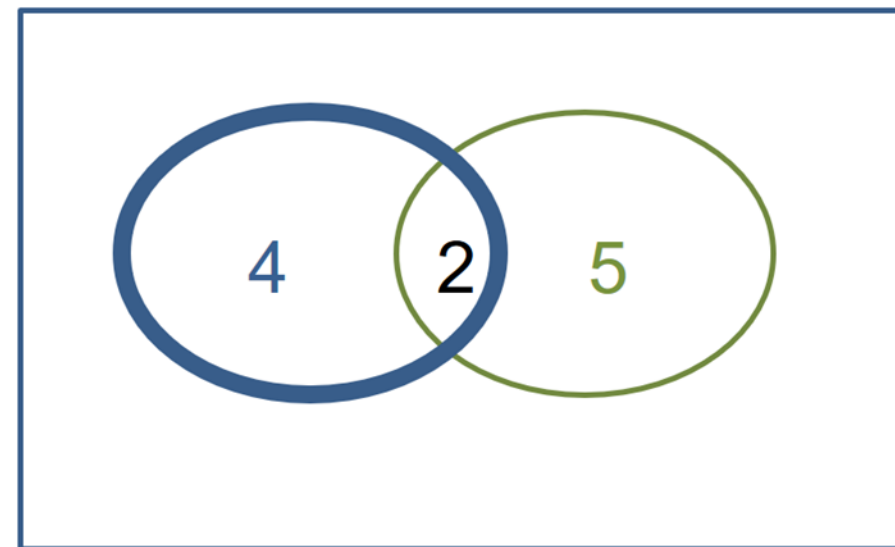
The context is very different!



Diagrams can help (7)

We are within the blue circle:

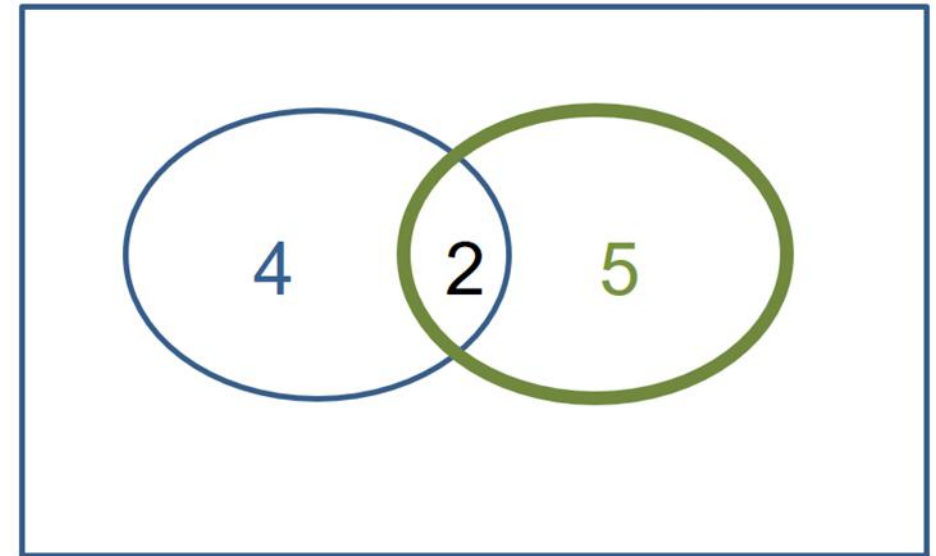
$$P(G|B) = \frac{2}{6}$$



Diagrams can help (8)

We are within the green circle:

$$P(B|G) = \frac{2}{7}$$



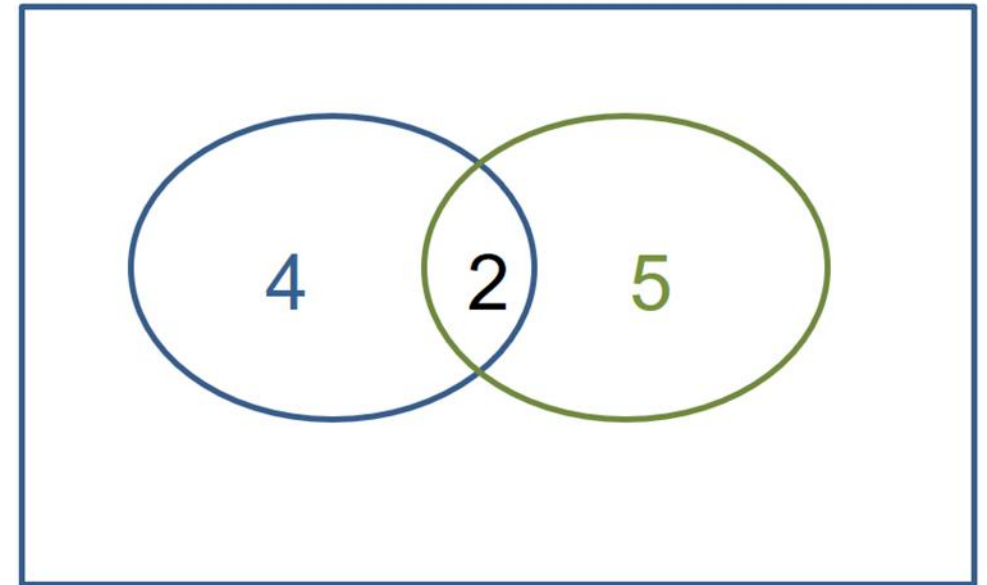
Bayes' Theorem - AND

A little rearranging of $P(G|B) = \frac{P(G \text{ and } B)}{P(B)}$
gives:

$$P(G \text{ and } B) = P(G|B)P(B)$$

If we had known the marble was Green and
wanted a Blue one, we would find:

$$P(B \text{ and } G) = P(B|G)P(G).$$



Bayes' Theorem - AND

Of course,

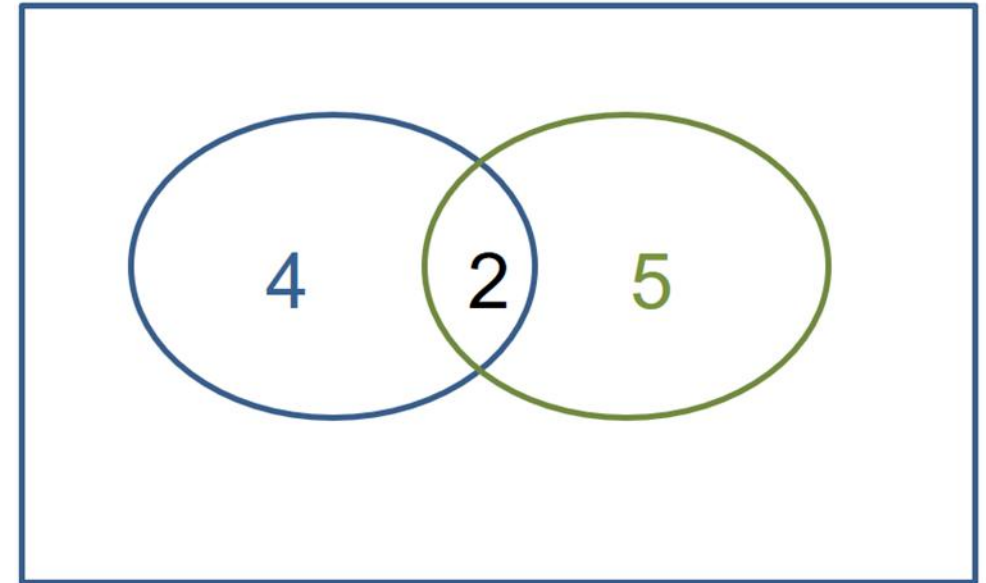
$$P(G \text{ and } B) = P(B \text{ and } G).$$

So:

$$P(G|B)P(B) = P(B|G)P(G).$$

We can reverse conditionalities!

*(This is **Bayes' Theorem.**)*



Law of total probability

Let **A** and **B** be collectively exhaustive events, then $P(A)+P(B)=1$

Let **C** be another event, then

$$P(C) = P(C \text{ and } A) + P(C \text{ and } B)$$

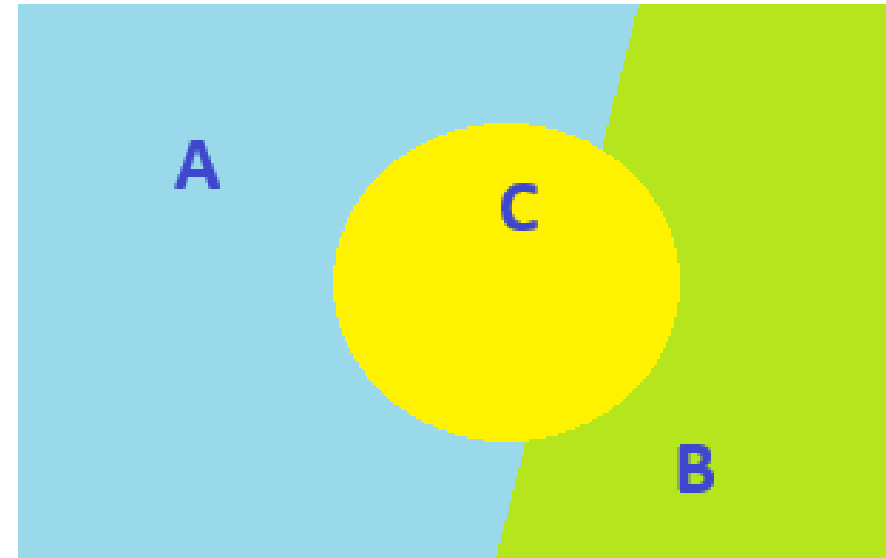
because A and B are exhaustive.

Given that

$$P(C \text{ and } A) = P(C|A)P(A) \text{ and } P(C \text{ and } B) = P(C|B)P(B)$$

then

$$P(C) = P(C|A)P(A) + P(C|B)P(B)$$



Probability (part 2)

Week 2

During the last session

- What are probability measures?
- How to calculate probabilities?
- How to combine probabilities?

During the last session

- **Probability** = (*fraction of desired outcomes*)/ N
- **Conditional Probability:** $P(A|B)$, same way to calculate but we restrict according to the given event
- **AND** $P(B \text{ and } A) = P(B|A)P(A)$
 - Bayes' rule: $P(A|B)P(B) = P(B|A)P(A)$
 - If A,B *independent*, then $P(A \text{ and } B) = P(A) \times P(B)$
- **OR** $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$
 - If A,B *mutually exclusive*, then $P(A \text{ and } B) = 0$ and $P(A \text{ or } B) = P(A) + P(B)$
- Law of total probability: if A and B are collectively exhaustive then
$$P(C) = P(C|A)P(A) + P(C|B)P(B)$$

Outline of the session

- **Bayes' theorem** and applications
- **Expectations**
- **Counting Arguments**

Bayes' theorem (rule or law)

$$P(A | B)P(B) = P(B | A)P(A)$$

Bayes' theorem allows to use new information to update (revise) some *prior* probability. The updated probability is called the *posterior* probability.

Example 1: The Faulty Circuit Board

Suppose a tech company gets its circuit boards from two suppliers: **Supplier A** and **Supplier B**.

- **Supplier A** provides **60%** of the boards.
- **Supplier B** provides **40%** of the boards.

- **Supplier A** is very reliable: only **5%** of their boards are faulty.
- **Supplier B** is less reliable: **20%** of their boards are faulty.

A technician picks a random board from the warehouse and tests it. **The board is faulty.** What is the probability that this specific faulty board came from **Supplier B**?

Example 1: The Faulty Circuit Board



Let **A** be the event 'the board is from Supplier A', and **B** be the event 'the board is from Supplier B'.
Let **F** be the event that the board is faulty.

Prior Probability, $P(B)$:

$P(B)$: The probability it's from Supplier B regardless of its condition. Since they provide 40% of the stock,

$$P(B) = 0.4$$

Also, the probability the board is from Supplier A (the complementary event). $P(A) = 0.6$

The Conditional Probabilities:

$P(F|B)$: The probability a board is faulty, *given* it came from Supplier B. We know this is 0.2

$P(F|A)$: The probability a board is faulty, *given* it came from Supplier A. This is 0.05

The Total Probability of a Faulty Board $P(F)$:

From the law of total probability, the overall chance of picking a faulty board from the whole warehouse is

$$P(F) = P(F|B)P(B) + P(F|A)P(A)$$

$$P(F) = 0.2 \times 0.4 + 0.05 \times 0.6 = 0.08 + 0.03 = 0.11$$

Example 1: The Faulty Circuit Board

The Posterior Probability $P(B | F)$:

Now we apply Bayes' Theorem to find the probability the board is from Supplier B, now that we *know* it is faulty:

$$P(B | F) = P(F | B)P(B) / P(F)$$
$$P(B | F) = 0.2 \times 0.4 / 0.11 = 0.727$$

Finally: There is a **0.727 probability** that the faulty board came from Supplier B.

Even though Supplier B only provides 40% of the total boards, their higher failure rate (20%) makes them the "prime suspect" when a defect is found.

Example 2: Credit Rating "Warning"



Imagine you are an analyst looking at a portfolio of corporate bonds. You want to know the probability that a company will Default (**D**) within a year if its credit rating is **Downgraded** (**W** for "Warning").

- **Prior Probability of Default:**

Historically, only 2% of companies in this sector default in a given year.

- **The "Accuracy" of a Downgrade:**

If a company is truly going to default, the rating agencies are good at spotting it: they downgrade 80% of those companies in advance.

However, they also issue "False Alarms" Of the companies that don't default, 10% of them still get downgraded anyway due to general market volatility.

A company in your portfolio has just been downgraded. What is the probability it will actually default?

Example 2: Credit Rating "Warning"

The Prior Probabilities:

- $P(D)$: The probability of default = 0.02
- $P(ND)$: The probability of not defaulting = 0.98

The Conditional Probabilities (The Evidence):

- $P(W|D)$: The probability of a downgrade given they will default = 0.80 (True Positive)
- $P(W|ND)$: The probability of a downgrade given they will not default = 0.10 (False Positive)

The Total Probability of a Downgrade, $P(W)$:

- We find the total chance of any company getting downgraded, whether they are actually failing or not

$$P(W) = P(W|D)P(D) + P(W|ND)P(ND)$$

$$P(W) = 0.80 \times 0.02 + 0.10 \times 0.98 = 0.016 + 0.098 = 0.114$$

- So, 11.4% of all companies will receive a downgrade

Example 2: Credit Rating "Warning"

- The **Posterior Probability**, $P(D | W)$:

Now we calculate the probability the company defaults, given that the downgrade has occurred

$$P(D | W) = P(W | D)P(D) / P(W)$$

$$P(D | W) = 0.80 \times 0.02 / 0.114 = 0.1404$$

- Hence, there is only a 0.1404 probability that the company will actually default.
- Even though a downgrade sounds scary and the agency is "80% accurate" at spotting failures, the overall rate of default is so low (2%) that a downgrade is still more likely to be a "false alarm" than a guaranteed bankruptcy

Bayes' Theorem

Bayes' theorem is a way of using new information to update some **prior probability**. The updated probability is called the **posterior probability**.

Expected value

Expected value: an Example

- A company is trying to decide between two investment portfolios.
- Let's call them the “**Optimistic Portfolio**” and the “**Pessimistic Portfolio**”.
- They both rely on the market's movements.
- Their outcomes are summarised in this table:

	Optimistic	Pessimistic
Market Up	+130	+50
Market Down	-150	-40

Expected value: an Example

We could just average the two scenarios:

$$(130 + (-150))/2 = -10$$

$$(50 + (-40))/2 = 5$$

	Optimistic	Pessimistic
Market Up	+130	+50
Market Down	-150	-40

...but this assumes the two Market situations are **equally likely**

Expected value: an Example

We could look at history to get an idea of the probabilities, or use some other, more sophisticated, method

0.6

0.4

	Optimistic Return (£)	Pessimistic Return (£)
Market Up	+130	+50
Market Down	-150	-40

We can still calculate the **weighted average** outcome

- $130 \times 0.6 + (-150) \times 0.4 = 18$
- $50 \times 0.6 + (-40) \times 0.4 = 14$

Expected value: an Example

- More formally: $E(X) = \sum P(X = x) \times x$
- The **expected** value (i.e. the **mean predicted value**) is the sum of the probabilities times the payoffs.
- Of course, there are other ways to decide:
 - Is the most likely event good or bad?
 - Do you maximise potential profit?
 - Do you minimise potential loss?
- **The expected value gives a good long-term view**

Counting Arguments:

Combinations and Permutations

Combinations and Permutations

- Problems frequently exist, where selections are made from groups of items.
- In these cases, it is useful to be able to **calculate the number of selections** that are possible,
 - If the **order** of selection **is important**, we use **permutations**.
 - If the **order** of selection **does not matter**, we use **combinations**.

Permutations (order is important)

- The number of permutations, where **r** items are to be selected from a **group of size n**, is given by the formula:

- **With** replacement/repetition – e.g. pin code

- $P_{(n,r)} = n^r$

- **Without** repetition – e.g. bookshelf

- ${}_nP_r = \frac{n!}{(n-r)!}$



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Permutations without replacement/repetition

Imagine you have **10 different books** and want to calculate how many possible ways you can arrange them on a bookshelf which **has a capacity for 5 books**. After you place the first book, the second book must be a different book.

$${}_nP_r = \frac{10!}{(10-5)!} = \frac{10!}{5!} = \frac{10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{5 \times 4 \times 3 \times 2 \times 1} = 30,240$$

Check if you can calculate it on your calculator.



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Combinations

(order is NOT important)

- **With repetition** – e.g. ice cream shop

- $$C_{(n,r)} = \frac{(r+n-1)!}{r!(n-1)!}$$

- **Without repetition** – e.g. lotto

- $${}^nC_r = \frac{n!}{r!(n-r)!}$$



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Combinations without repetition

- **What is the probability of winning the National Lottery (Lotto) jackpot?**
- To win the jackpot you must correctly guess the **six numbers drawn from the numbers 1 to 49**. These numbers can occur **in any order**, so you would use the **combination formula** for this calculation. In this case **n=49** and **r=6**, so:

$${}_nC_r = \frac{n!}{r!(n-r)!}$$

$${}_{49}C_6 = \frac{49!}{r!(n-r)!} = \frac{49!}{6! \times (49-6)!} = \frac{49 \times 48 \times 47 \times 46 \times 44 \times 43!}{6! \times 43!} = 13,983,816$$

- **This is nearly 14 million to 1!**

Combinations and Permutations

- Summary table

	Order matters Permutations	Order doesn't matter Combinations
No repetition/replacement	${}_nP_r = \frac{n!}{(n-r)!}$	${}_nC_r = \frac{n!}{r!(n-r)!}$
With repetition/replacement	$P_{(n,r)} = n^r$	$C_{(n,r)} = \frac{(r+n-1)!}{r!(n-1)!}$

This topic...

We have covered:

- **Probability** = (fraction of desired outcomes)/number of possible outcomes
- **Conditional Probability**: $P(A|B)$, same way to calculate but we restrict according to the given event
- **AND**: $P(A \text{ and } B) = P(A|B)P(B)$ and $P(B \text{ and } A) = P(B|A)P(A)$
- **Law of total probability**: $P(C) = P(C|A)P(A) + P(C|B)P(B)$
- **Bayes' rule**: $P(A|B)P(B) = P(B|A)P(A)$
- If A,B **independent**, then $P(A \text{ and } B) = P(B \text{ and } A) = P(A)P(B)$
- **OR**: $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$
- **Expected value**
- **Combinations and permutations**

Next topic

- We will be looking at **probability distributions**.
- What they do and how we use them.
- We'll be starting with two distributions:
 - **Bernoulli**
 - **Normal**
- These neatly match up with **dichotomous** and **continuous data** and lead nicely into future topics (We will also extend into the binomial).
- **Reading:** Oakshott, chapter 7 and Taylor, chapter 5.